

CMSE 801 - Day 20

Object oriented programming

Nov 16, 2022

General Course Announcements

- Homework #4 is due Wednesday Nov 16 at 11:59 p.m.
- Project's deadline Sunday Dec 4 at 11:59 p.m.
- Homework #5 is due Wednesday Nov 30 at 11:59 p.m.

Questions/comments from Day 20 PCA

- More examples of creating objects using composition / Could you give more examples about composition
- Can we please go over how to do the area in class? When I did use a for loop I kept getting errors, and the same thing happened with this Day 20 Pre-Class.
- Please go over the last question again in class / please go over the last part
- going over composition vs inheritance and how they are similar/different. An example or two would be incredibly helpful for me. / How would you describe when it is better to use inheritance or composition?
- Will it really be more simple to use composition? I mean when I want to use composition for class of zoo, I have to check the methods from animal class and search for the one I want for composition, but why not directly define the method ? It may be easier.
- I want to see where is my error in the last part of the code, to get the total area.
- I think creating objects are still kind of confusing to me.
- I think that it is more difficult to check the error in the inheritance and composition, because I really do not know which class the error comes from if all the classes are linked together.
- Just more time to focus on understanding object based coding and the differences between composition and inheritance.

```
In [1]: class Athlete():
        def __init__(self, name, sport, next_training, next_competition='NA'):
            ''' Initialize the object Building. '''
            self.name = name
            self.sport = sport
            self.next_training = next_training
            self.next_competition = next_competition
            return
        def set_next_competition(self, next_competition):
            self.next_competition = next_competition
            return
        def get_sport(self):
```

```

        return self.sport
    def get_info(self):
        s = 'Name: '+self.name+' Sport: '+self.sport+' N.T.: '+self.next_trainin
        return s

```

```

In [2]: class Boxer(Athlete):
        def __init__(self,name,next_training,numberMatches=0):
            super().__init__(name,'boxe',next_training)
            self.numberMatches = numberMatches
            return
        def get_numberMatches(self):
            return self.numberMatches

```

```

In [3]: Ali = Boxer('Ali','Nov 25')
        Ali1 = Athlete('Ali','Boxe','Nov 11')

```

```

In [4]: print(Ali.get_info())
        print(Ali.get_numberMatches())
        print(Ali1.get_info())
        print(Ali1.get_numberMatches)

```

Name: Ali Sport: boxe N.T.: Nov 25 N.C.: NA

0

Name: Ali Sport: Boxe N.T.: Nov 11 N.C.: NA

```

-----
AttributeError                                Traceback (most recent call last)
<ipython-input-4-0e166b7f6382> in <module>
      2 print(Ali.get_numberMatches())
      3 print(Ali1.get_info())
----> 4 print(Ali1.get_numberMatches)

```

AttributeError: 'Athlete' object has no attribute 'get_numberMatches'

```

In [5]: class Boxer_1():
        def __init__(self,name,next_training):
            self.sport = 'boxe'
            self.o = Athlete(name, self.sport, next_training)
        def get_info(self):
            return self.o.get_info()

```

```

In [6]: Ali2 = Boxer_1('ALI','DEC 1')
        print(Ali2.get_info())
        print(Ali2.sport)
        print(Ali2.o.next_competition)
        print(Ali2.next_competition)

```

Name: ALI Sport: boxe N.T.: DEC 1 N.C.: NA

boxe

NA

```

-----
AttributeError                                Traceback (most recent call last)
<ipython-input-6-b75c1269261c> in <module>
      3 print(Ali2.sport)
      4 print(Ali2.o.next_competition)
----> 5 print(Ali2.next_competition)

```

AttributeError: 'Boxer_1' object has no attribute 'next_competition'

```

In [7]: dic_athlete = {}
        dic_athlete['Ali'] = Ali

```

```
dic_athlete[1] = 'b'
```

```
In [8]: print(dic_athlete)
print(dic_athlete['Ali'])
print(dic_athlete['Ali'].get_info())
print(dic_athlete[1])
print(dic_athlete[0])
```

```
{'Ali': <__main__.Boxer object at 0x7fd7a84705e0>, 1: 'b'}
<__main__.Boxer object at 0x7fd7a84705e0>
Name: Ali Sport: boxe N.T.: Nov 25 N.C.: NA
b
```

```
-----
KeyError                                Traceback (most recent call last)
<ipython-input-8-f390df0b71b9> in <module>
      3 print(dic_athlete['Ali'].get_info())
      4 print(dic_athlete[1])
----> 5 print(dic_athlete[0])

KeyError: 0
```

```
In [ ]: # last part of the question 3.3
def get_animal_names(self, sort=False):
    """
    Return animal names in the ZOO.
    Return sorted names if parameters sorted==True
    """
    names = list(self.animals.keys())

    if sort == True:
        # we have to return sorted names; so sort the names
        names = sorted(names)

    return names

def total_area(self):
    """
    Return the total ZOO area to host all animals defined in the object.
    """
    area = 0

    #
    # add the content of this method; the method should return total area re
    # use the code from Day 19 In-class assignment - code for computing requ
    #
    for key in self.animals:
        area += self.animals[key].get_required_area()

    return area
```